



MCP (Model Context Protocol) Quiz

Total Questions: 20

Type: MCQ + Scenario-Based + Short Answer

Difficulty: Medium to Advanced



Section A: Multiple Choice Questions (MCQs)

Q1. What is the primary purpose of the Model Context Protocol (MCP)?

- A. Data storage
 - B. Standardized agent-tool communication
 - C. Model training
 - D. UI rendering
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Q2. Which MCP integration pattern focuses on delegating tasks across multiple agents?

- A. Single-agent pattern
 - B. Pipeline pattern
 - C. Multi-agent orchestration
 - D. Batch processing
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Q3. In MCP, tool invocation typically follows which structure?

- A. Direct function call
 - B. API request with structured schema
 - C. Manual input
 - D. File upload
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Q4. Which component ensures that agents can safely share data?

- A. Tokenization
 - B. Secure context sharing
 - C. Logging
 - D. Serialization
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Q5. What is a key advantage of MCP in enterprise environments?

- A. Faster UI rendering
 - B. Vendor lock-in
 - C. Interoperability across tools**
 - D. Reduced storage
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Q6. Which integration pattern is best suited for sequential task execution?

- A. Event-driven pattern
 - B. Pipeline pattern**
 - C. Mesh pattern
 - D. Parallel pattern
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Q7. What does “context” refer to in MCP?

- A. UI design
 - B. Model weights
 - C. Data passed between agents/tools**
 - D. Network latency
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Q8. Which of the following improves security in MCP?

- A. Hardcoding credentials
 - B. Context encryption**
 - C. Open access APIs
 - D. Public tokens
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Section B: Scenario-Based Questions

Q9. A financial company uses multiple AI agents for fraud detection, reporting, and alerting. Which MCP pattern is most suitable?

- A. Single-agent
 - B. Multi-agent orchestration**
 - C. Static pipeline
 - D. Batch processing
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Q10. An AI system fetches stock data, analyzes it, and generates a report in sequence. Which pattern is being used?

- A. Parallel processing
 - B. Pipeline pattern**
 - C. Mesh network
 - D. Event loop
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Q11. An agent calls an external weather API using MCP. What is this process called?

- A. Context sharing
 - B. Tool invocation**
 - C. Model tuning
 - D. Serialization
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Q12. A healthcare system ensures patient data is encrypted before sharing between agents. Which concept is applied?

- A. Data batching
 - B. Secure context sharing**
 - C. Logging
 - D. Caching
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Q13. A chatbot uses MCP to connect with CRM, database, and analytics tools. What is this an example of?

- A. Standalone AI
 - B. Enterprise agent integration**
 - C. Local processing
 - D. Static system
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Section C: Short Answer Questions

- **Q14.** Explain the difference between *pipeline pattern* and *multi-agent orchestration* in MCP.

Answer: Pipeline Pattern

- Linear flow (Step 1 → Step 2 → Step 3)
- Simple & predictable

- **Multi-Agent Orchestration**
 - Multiple agents collaborate dynamically
 - Flexible & intelligent decision-making

Key Difference:

Pipeline = fixed flow

Orchestration = dynamic collaboration

Q15. Describe how tool invocation works in MCP architecture.

Answer: Tool Invocation in MCP

- Agent identifies need (e.g., weather data)
- Creates structured request (JSON schema)
- Sends request via MCP
- Tool/API executes
- Response comes back as structured context
- Agent uses result for next step

Q16. What are the risks of insecure context sharing? Mention at least two.

Answer: Risks of Insecure Context Sharing

- Data leakage (sensitive info exposure)
- Unauthorized access
- Data tampering / manipulation
- Compliance issues (GDPR, HIPAA etc.)

Q17. How does MCP improve scalability in enterprise AI systems?

Answer: MCP improves scalability by

- Modular architecture (agents + tools separate)
 - Easy integration of new tools
 - Parallel agent execution
 - Reduced dependency coupling
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Section D: Case Study / Analytical

Q18.

A retail company wants to build an AI system that:

- Retrieves customer data
- Recommends products
- Sends personalized emails

Question:

Design an MCP-based integration approach and justify your chosen pattern.

Answer: Best Pattern: Multi-Agent Orchestration + Pipeline hybrid

Design:

1. Agent 1 → Fetch customer data
2. Agent 2 → Recommendation engine
3. Agent 3 → Email generator
4. Tool → Email API

MCP handles communication between all components

Justification:

- Personalization needs dynamic decisions → orchestration
- Sequential flow (data → recommend → email) → pipeline

Q19.

An organization is concerned about data privacy when multiple agents interact.

Question:

Suggest strategies using MCP to ensure secure context sharing.

Answer:

Secure Context Sharing Strategies

- Encryption (data in transit & at rest)
- Role-based access control (RBAC)
- Token-based authentication
- Context filtering (only required data share)
- Audit logging
- Secure APIs

Q20.

Compare MCP-based architecture with traditional API-based integration in enterprise systems.

Answer:

MCP (Model Context Protocol)-based architecture represents a modern, AI-native approach to system integration, whereas traditional API-based integration follows a more rigid, service-to-service communication model.

In traditional API-based systems, integration is typically point-to-point. Each service exposes endpoints, and developers must manually define how data is requested, formatted, and passed between systems. This often leads to tightly coupled architectures, where adding a new service or modifying an existing one requires significant changes across multiple components. Additionally, traditional APIs lack built-in context awareness, meaning each request is usually stateless and does not inherently carry rich, structured information about prior interactions.

In contrast, MCP-based architecture introduces a standardized protocol for communication between agents and tools. It is designed specifically for AI-driven environments, where multiple agents may need to collaborate, share context, and dynamically invoke tools. MCP enables structured context sharing, allowing agents to pass rich, meaningful data (such as intermediate results, user intent, or metadata) seamlessly across different components. This reduces redundancy and improves decision-making.

Another key difference lies in scalability and flexibility. MCP promotes a modular architecture where agents and tools are loosely coupled. New tools or agents can be integrated without disrupting the entire system, making it highly scalable for enterprise use. It also supports multi-agent orchestration, enabling intelligent coordination between different agents—something that is not natively supported in traditional API systems.

From a security perspective, MCP incorporates mechanisms like secure context sharing and controlled data access, which are essential in environments where sensitive information flows between multiple entities. Traditional APIs, while secure when properly implemented, require separate configurations for authentication, authorization, and data protection, often increasing complexity.

Overall, MCP-based architecture is more adaptable, context-aware, and suitable for complex AI ecosystems, while traditional API-based integration is more static, less flexible, and better suited for simpler, well-defined service interactions.
